MACHINE LEARNING / RESPONSIBLE AI / GOVERNANCE

When similar reports describe different AI harms

What text clustering can reveal - and what governance reviewers still need to judge.

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1,889	1,845	36
reports collected	provisionally retained	configurations explored*

The central finding

Clustering surfaced recognisable candidate themes, including fraud, surveillance, synthetic media and environmental resource concerns. The groups also reflected companies, application sectors and reporting language. Their overlap and instability limit how confidently they can guide decisions.

Practical implication: evaluate these groups as reading aids with human verification. The study does not demonstrate a validated harm taxonomy or an improvement in review performance.

*28 configurations in the main comparison and 8 in a wider-k extension. Source titles and summaries are not reproduced in this report.

RESEARCH REPORT / 01

A governance question, grounded in reported risks

Research question

To what extent can text clustering reveal recurring AI harm themes in OECD AIM reports, rather than simply grouping reports about the same company or news story?

Why this matters

Reviewers need to compare descriptions that may concern different systems, affected people and forms of harm. Shared vocabulary can help organise that reading, but it can also conceal differences that matter for safeguards and accountability. This study tests that tension rather than assuming useful categories will emerge.

A fixed quarter, with a traceable collection

The sample contains 1,889 unique reports from the OECD AI Incidents and Hazards Monitor (AIM), dated 1 July-30 September 2026. The collection manifest records retrieval on 23 October 2026. Report dates are not necessarily the dates of the underlying events. [1]

July	August	September	Total
581	599	709	1,889

The collection used 22 overlapping official Excel exports. Repeated IDs across batches were removed, and the final unique count matched the monitor snapshot. All 92 days were represented. These checks support completeness relative to that accessible snapshot, not worldwide coverage of AI harms.

Scope decisions remain provisional

Targeted close reading covered 181 summaries. Six reports were proposed for exclusion and 38 marked uncertain, leaving 1,845 for the main experiment. The remaining 1,708 summaries were not individually adjudicated. Nineteen possible reporting families covered 41 records; they include follow-ups and uncertain matches, not confirmed duplicate events.

The export is text-rich but context-limited

Every report has a title and summary. Summaries average about 47 words. The company field is entirely empty, and country is missing for 59 reports. The text may mix actual harm, plausible future harm and general commentary. No reliable incident/hazard or severity labels were available in the export. [1,2]

RESEARCH REPORT / 02

Transparent methods. Weak separation.

Titles and summaries were converted to TF-IDF word and phrase features. A second representation used 100-component latent semantic analysis (LSA), followed by normalisation. Existing concept tags and other metadata were excluded from the clustering features. LSA retained about 24.3% of original feature variance; no pretrained sentence-embedding model was tested.

Method	Main settings and role
K-means	k = 4, 6, 8, 10, 12; 20 initialisations; seed 42. Fits ordinary Euclidean K-means on TF-IDF or normalised LSA.
Average-linkage hierarchical	Same k grid on both representations; cosine distance. Offers a deterministic alternative but can produce uneven groups.
DBSCAN	LSA cosine distance; eps = .20, .35, .50, .65; minimum samples = 5 or 10. Explores local density and unassigned reports.

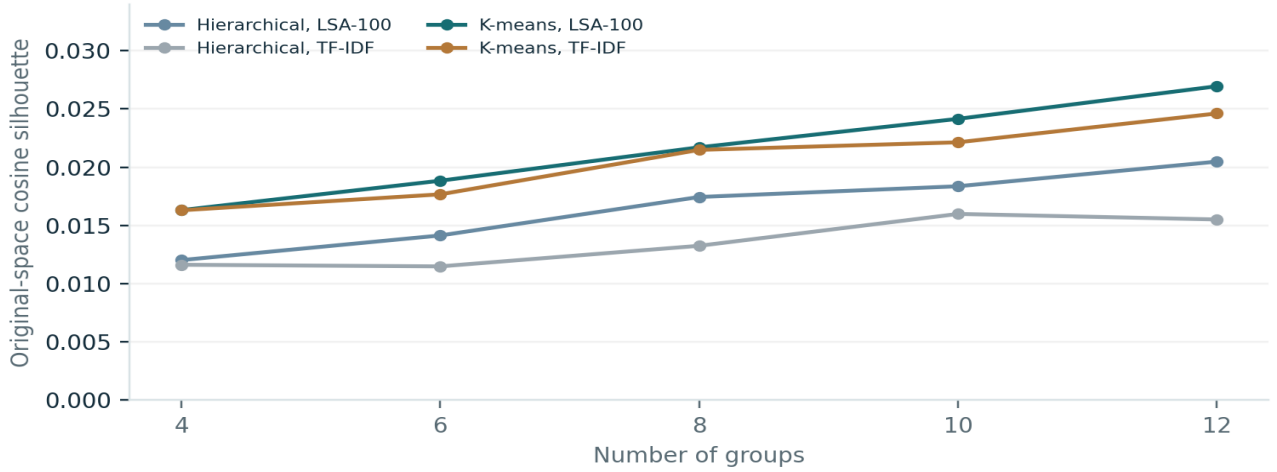


Figure 1. Cosine silhouette in the original TF-IDF space provides a common reference for full-coverage candidates. Values close to zero indicate substantial overlap; they are not accuracy scores.

The selected candidate

K-means on LSA-100, k = 12 achieved an original-space silhouette of **0.027** and a fitted-space score of **0.120**. About **21.1%** of reports had negative original-space silhouette. The operational selection rule required full coverage, 4-12 groups, at least 20 reports per group and no group above 45% of the corpus. These exploratory constraints were not externally validated.

At its tightest tested radius and minimum samples of 10, DBSCAN assigned only 6.7% of reports despite a higher silhouette on that subset. Coverage therefore matters alongside separation. The wider K-means extension reached 0.033 at k = 20; twelve groups are a practical ceiling in this study, not a demonstrated natural optimum.

RESEARCH REPORT / 03

Recognisable themes, with important boundaries

The selected partition produces twelve candidate reading queues. The names below are interpretations assigned after fitting, based on terms and central examples. They mix harm mechanisms, application areas, organisational contexts and broad commentary.

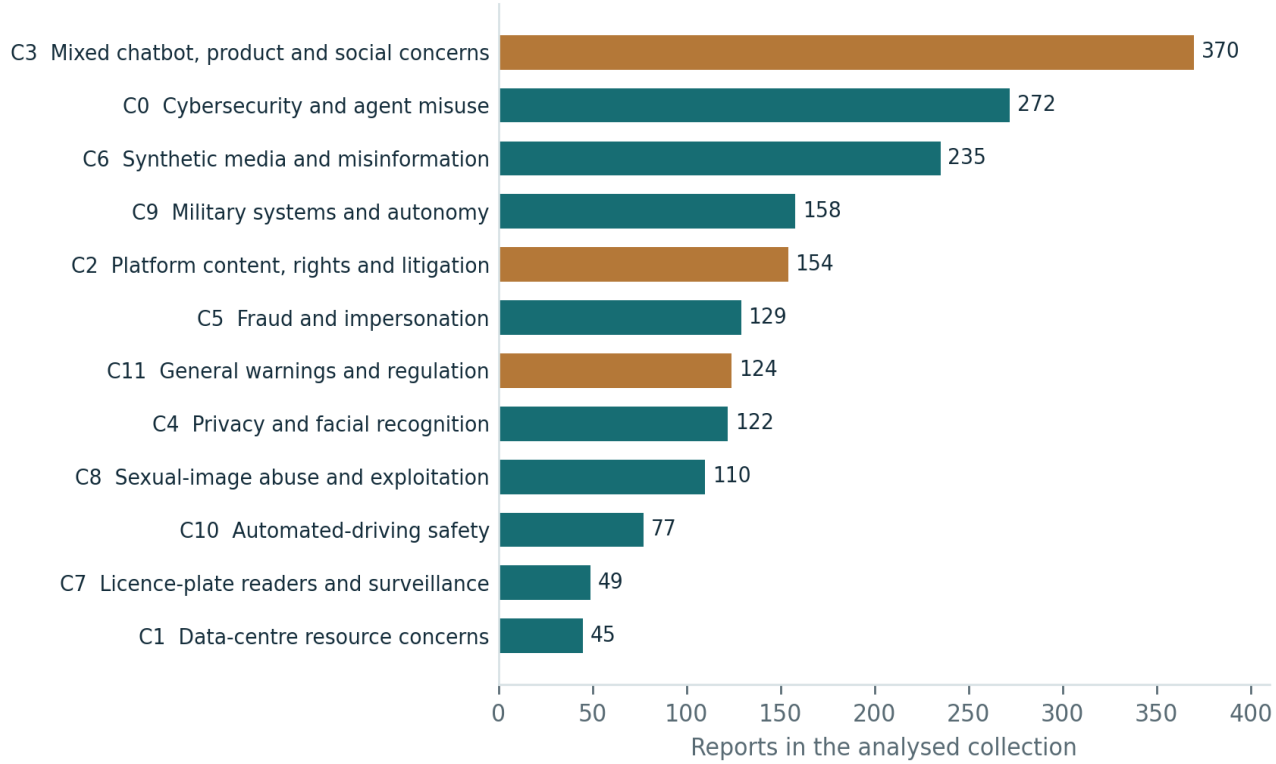


Figure 2. Counts describe reports in this collection, not independent events, real-world harm rates or severity. Gold highlights the three mixed groups examined on the next page.

Three observations

Related mechanisms can appear across separate groups. Fraud and misleading synthetic media share elements of deception, while privacy and licence-plate surveillance share questions about data collection and misuse. Different groupings could support different review questions.

A large group can be an unresolved remainder. The 370-report mixed chatbot, product and social-concern group is the largest. Its size does not establish a single dominant harm; its broad vocabulary signals a need for further triage.

Names and sectors still influence the result. About 42% of the cybersecurity queue mentions OpenAI and about 40% of the driving-safety queue mentions Tesla. These are overlapping mentions, not findings of responsibility. The ten-name detector is incomplete; it omits Flock, a prominent term in the licence-plate queue.

RESEARCH REPORT / 04

Inside the mixed groups

A shared label does not settle what belongs together. Contrasting descriptions within three existing clusters illustrate how different governance questions can be joined by similar language. These are candidate subthemes, not newly fitted subclusters or independently validated categories.

C2 / Platform content, rights and litigation / 154 reports

Child protection: terms such as children, minors and algorithms point towards safeguarding and platform moderation.

Creative rights: copyright, music and lawsuit point towards consent, licensing and compensation.

Interpretation: shared platforms and legal vocabulary can connect distinct affected groups and rights. Use this as an intake queue, then consider separate safeguarding and creative-rights tags. Retain both where a report spans them.

C3 / Mixed chatbot, product and social concerns / 370 reports

Employment effects: job, China and concerns connect descriptions of adoption and economic insecurity.

Reliance on outputs: chatbots, advice and legal connect descriptions of using unreliable outputs in sensitive tasks.

Interpretation: workforce transition requires different evidence from output validation and confidential-data handling. A broad product label can obscure that distinction. Audit a random sample before proposing finer categories.

C11 / General warnings and regulation / 124 reports

Broad future-risk warnings: existential, global and warns suggest long-horizon loss-of-control concerns.

Financial-system vulnerability: financial, potential and regulation suggest sector-specific preparedness questions.

Interpretation: distinguish a reported incident, a specific plausible hazard and general commentary. Record the harm pathway, evidence and time horizon. A warning is not proof that an adverse event occurred.

These readings are based on keywords and selected exported descriptions. Contrasting examples were checked against the saved assignments, but were not sampled randomly or independently verified. The two subthemes identified in each group do not exhaust its contents. No claim about their within-cluster prevalence is made.

RESEARCH REPORT / 05

A label can be readable while membership changes

Interpretability and stability answer different questions. A plausible name may describe central examples while other reports sit near a boundary. This study checked random starts, report subsets and alternative text inputs rather than relying on labels alone.

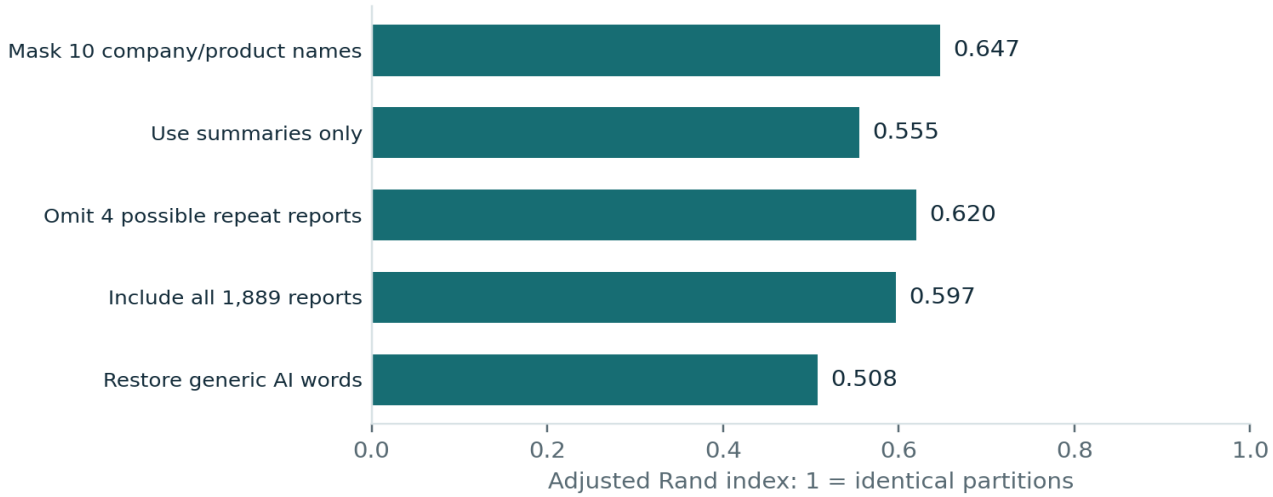


Figure 3. Each input-change experiment refits text features and the selected configuration. Adjusted Rand index (ARI) measures agreement between partitions on shared reports. It is not the percentage of reports that stayed in the same group.

Robustness check	Observed ARI range	What it assesses
Four extra random starts	0.573-0.671	Initialisation sensitivity on the same fitted representation.
Four 80% subsets; fit seed also varies	0.514-0.655	Combined subset and initialisation changes, with fixed features.
Four 80% subsets; fit seed fixed at 42	0.515-0.668	Conditional resampling stability without explicitly varying the numeric fit seed.

What these checks support

Masking ten names gives ARI 0.647; using summaries alone gives 0.555. Membership therefore depends on text choices. ARI alone does not tell us whether the new groups retain the same themes. Omitting four possible repeat reports gives 0.620, showing that refitting can amplify even a narrow input change.

The observed ranges overlap, but four runs per test do not isolate the causal contributions of sampling and initialisation. The subset checks keep the vocabulary and SVD fixed; they do not measure end-to-end feature stability. A fixed numeric seed also does not guarantee identical starting centres on different subsets.

RESEARCH REPORT / 06

An impact brief for governance teams

Decision supported now

Run a small supervised pilot to assess whether the candidate queues help reviewers organise reading. Human verification remains necessary before a report or theme informs a decision.

Candidate theme	Question or action to test	Evidence needed first
Fraud and impersonation	Add impersonation scenarios to staff awareness and incident intake.	Verify mechanisms and distinguish repeat reporting from separate events.
Privacy and surveillance	Ask who can access data, how long it is retained and how it could be misused.	Check original sources and the context of affected people.
Sexual-image abuse	Include safeguarding and abuse-response questions in reviews.	Use trained reviewers and verify accuracy and relevance.
Data-centre resources	Request energy and water evidence alongside claimed benefits.	Use measured local impacts; news volume is not an environmental footprint.

Who should do what

Governance teams: assign responsibility for checking scope and sources, keep uncertain reports visible and allow more than one theme per report. Do not derive compliance, responsibility or severity from cluster membership.

Policy teams: use verified examples to broaden consultations and scenario planning. Include smaller queues, and combine reporting data with evidence about exposure, affected people and severity before allocating resources.

Researchers: complete relevance and event checks, then ask two independent reviewers to assess random reports from every queue. Compare assisted and ordinary reading using review time, relevant cases missed and disagreements.

Limits to carry into any decision

Media coverage is selective; actual and potential harms are mixed; summaries may be inaccurate; reports may concern the same event. Relevance screening and independent thematic validation remain incomplete. No reduction in review time or harm has been demonstrated. The study does not estimate real-world prevalence, establish legal breaches or measure unequal exposure.

RESEARCH REPORT / 07

What this investigation contributes

A partial answer to the research question

Text clustering exposes candidate themes that could organise reading, but the tested pipeline does not demonstrate stable harm categories independent of companies and news topics. That boundary is the substantive finding: useful-looking labels need evidence before they guide action.

Lessons from the research process

Framing changes evaluation. Asking about governance usefulness required coverage, sensitivity and interpretability alongside a separation score.

Representation changes the picture. LSA improved compactness in a reduced space, but original-space separation remained weak. A cleaner plot is insufficient evidence of better harm categories.

Small or rare groups still matter. Filtering uncommon vocabulary can suppress unusual harms. Reporting volume should not become a proxy for importance.

Next steps

Complete scope and event adjudication; conduct an independent coherence audit; compare a pinned sentence-embedding model under the same protocol; and evaluate another quarter. Any workflow pilot should measure benefits and missed cases rather than assuming efficiency gains.

Reproducibility and attribution

The companion notebook contains the code, experiment settings, saved outputs and detailed limitations. Its saved aggregates can regenerate figures; full clustering recomputation requires the exact separately held source CSV. Python and analytical package versions are documented in the project. AI tools supported the preparation of this project. I reviewed the final analysis and report.

Sources and companion material

[1] OECD.AI. AI Incidents and Hazards Monitor. Source of the fixed-quarter collection; manifest retrieval date: 23 October 2026.

[2] OECD.AI. Overview, methodology and transparency disclosures. Context for incidents, hazards and media-based reporting. Referenced in the notebook on 30 October 2026.

[3] Barbuceanu, S. *Discovering recurring AI risk themes in OECD AIM reports.* Final reviewed capstone notebook, main experiment table, extended experiment results and aggregate profiles. Numerical source for all figures and findings in this report.

[4] Companion GitHub repository: ai-harm-text-clustering. Code and documented analytical workflow.

This report summarises analytical results and interpretations. Original exported titles and summaries are omitted. The notebook records an unresolved source-text reuse enquiry; this report does not grant rights over the underlying collection.